Partitional Primes

Definition, provability, and a finiteness heuristic

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A number class invented by Oliver Lau, together with the heuristic argument that it almost certainly contains only **finitely many** primes. This note records the definition, why these primes are cheap to prove, and the probabilistic budget that predicts their scarcity — so it does not get lost.

1 Definition

With $p(k)$ the integer **partition function**

$$p(0) = 1,\ p(1) = 1,\ p(2) = 2,\ p(3) = 3,\ p(4) = 5,\ p(5) = 7,\ p(6) = 11,\ \ldots$$

define the *partitorial*

$$\Pi(n) \;=\; \prod_{k=0}^{n} p(k) \;=\; p(0)\cdot p(1)\cdot p(2)\cdots p(n),$$

and the **Partitional numbers** (operator $\vdash$, ASCII fallback §):

$$n \vdash = \;\Pi(n) \pm 1.$$

Example ($n = 10$).

$$\Pi(10) = 1\cdot 1\cdot 2\cdot 3\cdot 5\cdot 7\cdot 11\cdot 15\cdot 22\cdot 30\cdot 42 = 960\,498\,000,$$

so the two candidates are 960 498 001 and 960 497 999.

How fast does $\Pi(n)$ grow?

$\Pi(n)$ grows *much* faster than the factorial. The cleanest way to see this is not to look at the number itself — it quickly has thousands of digits — but at its **logarithm**. A handy fact: the number of decimal digits of a whole number X is about $\ln X/\ln 10 \approx \ln X/2.303$. So "how big is $\Pi(n)$" really means "how big is $\ln\Pi(n)$," and that is the quantity we track.

Step 1 — turn the product into a sum. The logarithm turns multiplication into addition, which is exactly what we want for a product:

$$\ln\Pi(n) \;=\; \ln\prod_{k=0}^{n} p(k) \;=\; \sum_{k=0}^{n} \ln p(k).$$

So the size of $\Pi(n)$ is just the running total of the sizes of the individual partition numbers $p(k)$.

Step 2 — how big is a single partition number? Here we borrow one famous result, the *Hardy–Ramanujan formula* (1918): the partition numbers themselves already grow almost exponentially, and their logarithm behaves like

$$\ln p(k) \;\sim\; \pi\sqrt{\tfrac{2}{3}\,k} \;\approx\; 2.565\,\sqrt{k}.$$

In words: each partition number is roughly "e to the power of (a constant times $\sqrt{k}$)." This is the only outside ingredient we need — everything else is bookkeeping.

Step 3 — add them all up. Summing Step 2 over $k = 0, \ldots, n$ means adding up $\sqrt{0} + \sqrt{1} + \cdots + \sqrt{n}$. A sum of $\sqrt{k}$ is well approximated by the area under the curve $\sqrt{x}$, i.e. its integral:

$$\sum_{k \le n} \sqrt{k} \;\approx\; \int_0^n \sqrt{x}\,\mathrm{d}x \;=\; \tfrac{2}{3}\, n^{3/2}.$$

Putting Steps 2 and 3 together (the constant is $\pi\sqrt{2/3} \cdot \frac{2}{3} \approx 2.565 \cdot 0.667 \approx 1.71$):

$$\boxed{\ln \Pi(n) \;\sim\; 1.71 \cdot n^{3/2}}$$

What this means in plain terms. The exponent is $n^{3/2} = n\sqrt{n}$, i.e. "n times the square root of n." That one square-root factor is what makes Partitional numbers explode faster than factorials: for the factorial, $\ln(n!) \sim n \ln n$ grows only like "n times the (slow) logarithm of n," whereas here it grows like "n times the (much larger) $\sqrt{n}$." Concretely, $\ln \Pi(n) \approx 1.71\, n^{3/2}$ translates via Step 1's digit rule (digits $\approx \ln \Pi(n)/2.303$) into

$$\text{digits of } \Pi(n) \;\approx\; 0.74 \cdot n^{3/2}.$$

Check it: at $n = 316$ this predicts $0.74 \cdot 316^{3/2} \approx 4160$ — the right ballpark for the actual 3255 digits (the formula is an asymptotic, a bit generous at "small" n). At $n = 1000$ it is already about 20 000 digits, and at $n = 10\,000$ roughly 740 000. This breakneck growth is precisely why Partitional primes thin out so fast — see the finiteness heuristic in Section 3.

2 Why these primes are cheap to prove

Two kinds of "is it prime?" There is a fast, cheap test (a *primality test*) that says "this number is almost certainly prime" — good enough to go hunting with, but not a proof. And there is the harder job of a watertight **proof** that a number really is prime, with no "almost." For a number with thousands of digits, how expensive that proof is depends entirely on *what we know about the number's neighbours*, $N - 1$ and $N + 1$.

The expensive route: ECPP. For a "random" large prime — one with no special structure — the only general proof method is **ECPP** (Elliptic Curve Primality Proving). It works, but it is slow and hungry for memory: proving a few-thousand-digit number can take hours and gigabytes of RAM. This is the wall the rest of this project keeps running into.

The cheap route: knowing the factors of a neighbour. There is a classical shortcut, more than a century old. *If* you already know the complete prime factorization of $N - 1$ *or* of $N + 1$, then you can **prove** N prime quickly and with almost no memory. The catch is normally the "if": factoring a big number is itself hopeless.

Why factoring $\Pi(n)$ costs us nothing. The "if" is normally the whole problem — but for Partitional numbers it is free, and the reason is a small, almost childish observation: **the prime factorization of a *product* is just the factorizations of its pieces, thrown together.** And $\Pi(n)$ *is* a product — one we assembled ourselves out of the small partition numbers $p(0), \ldots, p(n)$. We never set a factoring algorithm loose on the giant $\Pi(n)$; we factor each little $p(k)$ (they have only a handful of digits) and collect. For $n = 10$:

$$\Pi(10) = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot \underbrace{15}_{3\cdot5} \cdot \underbrace{22}_{2\cdot11} \cdot \underbrace{30}_{2\cdot3\cdot5} \cdot \underbrace{42}_{2\cdot3\cdot7} \;=\; 2^4 \cdot 3^4 \cdot 5^3 \cdot 7^2 \cdot 11^2 \;=\; 960\,498\,000.$$

We have just read off the complete factorization of 960 498 000 without ever treating that nine-digit number as a monolith. The identical thing happens at $n = 316$: the product has 3255 digits, yet its factorization is handed to us for free, because every building block is tiny.

The trick, run in reverse. It is worth savouring how backwards this is. Normally you are handed a number N and get stuck, because *its* neighbours $N-1$ and $N+1$ both refuse to factor. Here we go the other way round: we *start* from a number we have already factored completely — $\Pi(n)$, by construction — and ask which nearby numbers it now lets us certify. We are not picking a target and then hunting for its factors; we are picking targets *because* their factored neighbour is already in hand. And a fully factored $\Pi(n)$ has exactly two immediate neighbours to offer: $\Pi(n)+1$ and $\Pi(n)-1$.

Two candidates, two directions. Because $\Pi(n)$ sits right between those two candidates, the one free factorization serves both of them — once on each side. Take the first variant, $N=\Pi(n)+1$: its *lower* neighbour is $N-1=\Pi(n)$, precisely the number we hold in factored form, and knowing the factors of $N-1$ is exactly what the classical $N-1$ **test** wants (Pocklington's theorem, descended from Lucas and later sharpened into the Brillhart–Lehmer–Selfridge, or "BLS", form). The second variant, $N=\Pi(n)-1$, tilts the same fact the other way: now the factored number is the *upper* neighbour, $N+1=\Pi(n)$, which is the input for the mirror-image $N+1$ **test** (built on Lucas sequences, likewise in a BLS form). So the single pre-factored number $\Pi(n)$ plays the part of "$N-1$" for one candidate and "$N+1$" for the other: two sides of the same coin, and between them they cover both Partitional variants.

The idea behind the two tests, without formulae

Both tests rest on one picture: a prime N forces a certain *cycle* to have an exact length, and you could only have built a cycle of that length if N really is prime. Checking the length precisely is where the factors come in.

The $N-1$ test. Picture a clock face whose positions are the numbers $1, 2, \ldots, N-1$. Pick a starting value a and "tick" by repeatedly multiplying by a, always wrapping around (that is, keeping the remainder after dividing by N). Sooner or later the hand lands back on its start; the number of ticks that takes is its *period*. A classical fact (Fermat's little theorem) says that when N is prime, some choice of a runs with the *full* period $N-1$ — the hand tours every position before returning home. Turn that around and it becomes a proof: if you can *exhibit* an a whose period is exactly $N-1$, then the clock truly has $N-1$ distinct positions, and that happens only when N is prime. The delicate part is confirming the period is the *whole* $N-1$ and not some shorter fraction of it — and that is where you need the prime factors of $N-1$: for each prime q dividing $N-1$, you check that the hand is *not* yet home after only $(N-1)/q$ ticks. No factors of $N-1$, no check. A pre-factored $N-1$ is the golden ticket.

The $N+1$ test. When it is $N+1$ you can factor instead, you play the very same game on a *larger* clock. To build it you widen the number system a little: roughly, you admit one new "imaginary" quantity — a square root that simply does not exist among the ordinary remainders — much as the complex numbers are made by bolting $\sqrt{-1}$ onto the reals. In that enlarged world a prime N conjures up a clock of size $N+1$, and the role of "keep multiplying by a" is taken over by a Fibonacci-like recipe (technically a *Lucas sequence*: each term is a fixed blend of the two before it). The rest is identical: exhibit such a sequence whose period is the full $N+1$, confirm it does not collapse early at $(N+1)/q$ for each prime q dividing $N+1$, and N stands proven. Same idea, mirror side — and this time it is the factors of $N+1$ that you must know.

Why this is so cheap. Each check above is one "run the hand to a certain number of ticks" — in arithmetic terms, one modular exponentiation — per prime factor of $\Pi(n)$. That is fast, and it uses only $O(N)$ memory (storage proportional to the number's own length, nothing worse). It scales far beyond what ECPP can reach: we proved $\Pi(316)-1$ (3255 digits) deterministically in

∼9 minutes on a laptop via the $N+1$ method, where plain ECPP on the same number would take hours and heavy RAM.

The same trick in the wild. Partitional numbers are not the only "product ±1" family, and the others are hunted in earnest. The factorial primes $n! \pm 1$, the primorial primes $p\# \pm 1$, and — since 2025 — the compositorial primes $n!/n\# \pm 1$ (here $n\#$ is the primorial, so $n!/n\#$ is the product of all *composite* numbers up to n) are all searched by the volunteer project **PrimeGrid**. Every record they report is a genuine, fully *proven* prime of hundreds of thousands of digits — not a mere probable prime — and the reason is exactly the mechanism above: each of these forms carries its own pre-factored neighbour, so the $N \pm 1$ proof stays cheap at sizes where ECPP would be hopeless. Partitional numbers share that machinery precisely; what sets them apart is only their faster growth (Section 3), which alone pushes them onto the rare "finitely many" side.

(Fine print: for very large n a few of the $p(k)$ themselves become big and might not factor quickly. The BLS method still works as long as the part we did *factor covers more than the cube root of N — which in practice holds comfortably — with ECPP needed only on whatever small leftover remains. This has not been necessary in the range explored so far.)*

3 Why there are probably only finitely many

This is not a proof (none is possible — see Section 4). It is a back-of-the-envelope argument that nonetheless pins the answer down surprisingly well.

Step 1 — how likely is one candidate to be prime? A deep but standard result (the *prime number theorem*) says: among numbers near a large value X, roughly one in every $\ln X$ is prime. So if you hand me a number of size X and I know nothing special about it, my best guess is that it is prime with probability about $1/\ln X$. For our candidate $\Pi(n) \pm 1$, that size is $X = \Pi(n)$, so

$$\Pr[\,\Pi(n) \pm 1 \text{ prime}\,] \;\approx\; \frac{C}{\ln \Pi(n)}.$$

The fudge factor C just bundles two real effects: we get two shots at each n (the $+1$ and the -1), and these candidates are known to have *no small factors* (they were built that way), which makes them a bit likelier to be prime than a truly random number. We will pin C down from the data in a moment.

Step 2 — add up the chances over all n. The expected *total* number of Partitional primes is just the sum of those per-n chances. And now the growth rate from Section 1 does all the work. Because $\ln \Pi(n) \approx 1.71\, n^{3/2}$, each term shrinks like $1/n^{3/2}$:

$$E_{\text{total}} \;=\; \sum_{n \geq 2} \Pr[\,\Pi(n) \pm 1 \text{ prime}\,] \;\approx\; C \cdot \sum_{n \geq 2} \frac{2}{\ln \Pi(n)} \;\propto\; \sum_{n \geq 2} \frac{1}{n^{3/2}}.$$

Step 3 — does that sum add up to something finite? This is the crux, and it is a schoolbook distinction. An infinite sum of ever-smaller terms can either pile up without limit (*diverges*) or settle on a finite total (*converges*). The textbook dividing line is the exponent on n: $\sum 1/n^1$ diverges (grows forever), while $\sum 1/n^s$ for any exponent $s > 1$ converges. Our exponent is $s = \frac{3}{2} > 1$, so

$$\sum_{n \geq 2} \frac{1}{n^{3/2}} \;\textbf{converges} \;\Longrightarrow\; E_{\text{total}} \text{ is } \textbf{finite}.$$

In plain words: the candidates grow so explosively (that $\sqrt{n}$ in the exponent again) that the chance of being prime drops away fast enough for the grand total of expected primes to be a finite number — not infinity.

Step 4 — put a number on it. We can fix C by matching the prediction to reality. Up to $n = 316$ we have found 19 primes, and the corresponding sum works out to $2\sum_{n\le 316} 1/\ln \Pi(n) = 6.89$. Setting $C \cdot 6.89 = 19$ gives $C \approx 2.76$. Carrying that same C out to *all* n (the tail of the sum adds very little) gives the **total lifetime budget**:

$$E_{\text{total}} \;\approx\; 2.76 \cdot 2\sum_{n\ge 2} \frac{1}{\ln \Pi(n)} \;\approx\; 2.76 \cdot 7.05 \;\approx\; \boxed{\text{19.5 primes — over } \textit{all } n.}$$

We have already found 19. **The budget is essentially spent.** The table below splits the remaining ∼0.5 of a prime across ranges of n ("bands"). E is the expected number of hits in that band; $P(\ge 1) = 1 - e^{-E}$ converts that into the probability of finding *at least one* (if hits land at random with average count E, the chance of drawing a blank is e^{-E}, so the chance of one or more is $1 - e^{-E}$):

Band	**expected** E	$P(\ge 1)$
(316, 920]	0.18	∼17%
(920, 1000]	0.01	∼1%
(1000, 2000]	0.07	∼7%
(2000, 4000]	0.05	∼5%
(4000, 40000]	0.07	∼7%
beyond 920, over all n, forever	**≈ 0.23**	**∼21%**

So the expected number of Partitional primes *beyond* the ones we already have is well under one: **most likely there are no more, at any n — ever.** Two honest caveats. First, C was fitted from only 19 data points at small n, so read the individual percentages as rough orders of magnitude, not precise odds. Second — and this is the robust part — the headline conclusion (expected count much less than 1 from here on) rides only on the growth rate $\ln \Pi(n) \sim n^{3/2}$, not on the exact value of C. Even if C were off by a factor of two, the answer would still be "probably none left."

Where this sits among "generator" primes

It helps to compare Partitional numbers with the other classic recipes that build candidates from a growing product or formula. The single thing that decides "finitely many vs. infinitely many" is whether the sum from Step 3 converges — which in turn is decided by how fast the numbers grow:

Family	**size** $\ln$**(value) grows like**	**the sum** $\sum 1/\ln$	**expectation**
Primorial $p\# \pm 1$	$n \ln n$	diverges	infinitely many
Factorial $n! \pm 1$	$n \ln n$	diverges	infinitely many
Compositorial $n!/n\# \pm 1$	$n \ln n$	diverges	infinitely many
Partitional $\Pi(n) \pm 1$	$\mathbf{n^{3/2}}$	**converges**	**finitely many**
Fermat $2^{2^k} + 1$	2^k	converges (fast)	finitely many

The primorial ($p\#$, the product of all primes up to p), the factorial, and the compositorial ($n!/n\#$, the product of all *composite* numbers up to n) all grow only moderately, their sums never settle, and all three are expected to yield infinitely many primes. The Fermat numbers grow so savagely (a tower of powers of 2) that their sum converges almost instantly — and indeed only 5 Fermat primes are known, with most experts expecting no more. Partitional numbers land on the same "finite" side as Fermat, just with a more generous budget (∼19 expected, versus the 5 Fermat primes). That is the company this new class keeps — and, among the product-based families, it is the lone outlier on the finite side.

4 Can it be proven or disproven?

Short answer: **no — not with today's mathematics**, in either direction. The heuristic above tells us what to *expect*; it does not settle anything. This is the same frustrating situation as with the Fermat primes, where "are there finitely many?" has been open for centuries.

- **We cannot prove there are finitely many.** To do that, you would need to show that from some point on, *every* $\Pi(n) \pm 1$ is composite (non-prime). But there is no mechanism forcing that. The usual way a formula gets "poisoned" into always being composite is a fixed small divisor — and these candidates are immune to exactly that. Here is why: the moment some prime q shows up as a factor of one partition number $p(k)$, it is a factor of the whole product $\Pi(n)$ for every $n \geq k$. So $\Pi(n)$ is a multiple of q, which means $\Pi(n) \pm 1$ is *one away* from a multiple of q and therefore *never* divisible by q. In other words the candidates are, by construction, coprime to (share no factor with) every prime that lives inside $\Pi(n)$ — they are designed to slip past all the small divisors. Ruling out primality forever would mean proving that an incredibly thin set of numbers eventually contains none at all, and no technique can do that.
- **We cannot prove there are infinitely many either.** That is hard even for the "easy" families whose sums diverge (primorial, factorial), where everyone believes the primes go on forever but nobody can prove it. For a convergent family like ours it is only more hopeless.

The one escape hatch that would turn this into a real theorem is an *algebraic* surprise: if $\Pi(n) \pm 1$ secretly always factored by a formula — the way $n^4 + 4$ always splits as $(n^2 - 2n + 2)(n^2 + 2n + 2)$ and so is composite for every $n > 1$ — then finiteness (in fact, "almost none") would follow instantly. Nothing hints at such a hidden factorization, and the coprimality argument above makes a *simple* one unlikely. But that identity, if it existed, is the only known route to an actual proof.

5 Known Partitional primes ($n \leq 320$, all proven)

n	variant	digits	method
2	$\Pi(n)+1$	1	$N-1$
3	$\Pi(n)+1$	1	$N-1$
3	$\Pi(n)-1$	1	$N+1$
4	$\Pi(n)+1$	2	$N-1$
4	$\Pi(n)-1$	2	$N+1$
5	$\Pi(n)+1$	3	$N-1$
6	$\Pi(n)+1$	4	$N-1$
6	$\Pi(n)-1$	4	$N+1$
7	$\Pi(n)+1$	5	$N-1$
7	$\Pi(n)-1$	5	$N+1$
10	$\Pi(n)+1$	9	$N-1$
13	$\Pi(n)+1$	15	$N-1$
14	$\Pi(n)-1$	17	$N+1$
18	$\Pi(n)+1$	27	$N-1$
22	$\Pi(n)+1$	38	$N-1$
67	$\Pi(n)+1$	258	$N-1$
198	$\Pi(n)-1$	1535	$N+1$
217	$\Pi(n)-1$	1780	$N+1$
316	$\Pi(n)-1$	3255	$N+1$

19 primes total (11 of form $\Pi(n)+1$, 8 of form $\Pi(n)-1$). No Partitional prime of either variant exists for $68 \leq n \leq 320$ except $n = 198, 217, 316$ — consistent with the heuristic's thinning tail.

6 Tools in this repo

- `partitional.gp` — explorer: `scan(nmax)` (table), `prove(n)` (both variants, deterministic, with timing), `report(nmax)` (list primes). The $N-1$ and $N+1$ BLS provers are validated against PARI's `isprime` for $n \leq 50$.
- `hunt_partitional.sh` — resumable, `nice`-d PRP search over a range, appending hits to `results/partitional_prp.log`; prove hits later with `gp -q partitional.gp` → `prove(n)`.